Quantitative Assessment of TA-ERT Cognitive Data: Reconstructing Effect Size, Longitudinal Velocity of Decline, and Point-Pool Attrition.
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The current investment thesis for Spruce Biosciences rests on the reality that efficacy of TA-ERT in reducing the HS-NRE biomarker is validated by consensus, but the FDA has been explicit: biomarker reduction must correlate with tangible clinical outcomes. I chose cognition as the primary clinical outcome of focus in this paper as MPS III-B is a neurodegenerative disorder. Thus, I assume BSID-III cognition is the "golden goose" of TA-ERT's clinical endpoints that will determine if HS-NRE is truly "reasonably likely to predict clinical benefit".

My goal is to look for the ground truth of the 2026 WORLDSymposium data and strip away all of the sponsor's modeled smoothing (used to obtain the p-values in the cognition table). I am specifically interested in seeing whether the data shows a genuine bending of the curve.

My analysis of the BSID-III raw scores identifies three phases. I would appreciate feedback on the biological plausibility of this thesis.
From Ages 4.0 to 7.0, velocity differences swing wildly. I view this as not a failure of TA-ERT, but a reflection of the developmental plateau in natural history. The untreated cohort has not yet hit the neurodegenerative cliff, making it mathematically impossible to show superiority in a noisy, small-sample environment.
From Ages 7.5 to 9.0, the signal of drug efficacy appears. As the untreated cohort enters a decline, the TA-ERT cohort appears to demonstrate a limited decline, perhaps stabilization. This 18 month window is the most plausible evidence that the drug intercepts the disease mechanism before biological debt leads to total cognitive collapse.
Beyond Age 9, the natural history data becomes plagued by survivorship bias. As the most severely affected untreated children reach the floor or drop out of testing, the remaining healthier survivors artificially pull the mean velocity upward. Recognizing this bias is critical to understanding why the treatment effect appears to narrow in late-stage data (I explain this later on).

To be objective and counter my own confirmation bias, I have conducted a statistical analysis of the BSID-III raw score data from the 2026 WORLDSymposium poster. These include Variance Reconstruction (reverse-calculating the standard deviations to expose the massive underlying heterogeneity), Hedges' g Correction (adjusting Cohen's d effect sizes to penalize for small sample sizes and unequal variances), Velocity Differential (calculating the first derivative of cognitive scores to isolate the "bending of the curve" and understand the extent of disease modification), and Point-Pool Attrition (calculating the implied shift in score per change in N).

To evaluate the clinical signal, the standard deviations of mean scores were reverse-calculated in order to expose the raw distribution of the efficacy populations. Standard Errors are common reporting metrics but can be deceptive in small-sample clinical trials (n<30) because it inherently shrinks as the sample size increases (SE=SD/root(N)).

Chronological Age	Study 902 BSID Mean Raw Score	Study 902 N at Risk	Study 902 Standard Error	Study 902 Standard Deviation	Study 201/Ext BSID Mean Raw Score	Study 201/Ext N at Risk	Study 201/Ext Standard Error	Study 201/Ext Standard Deviation	P-Value (Study 902 vs. Study 201/Ext)
4.0	42.98	5	10.8	24.1495	59.8	8	20	56.5685	0.111
4.5	47.3	6	9.9	24.2499	53.9	10	24.8	78.4245	0.55
5.0	48.1	10	12.5	39.5285	63	12	22.7	78.6351	0.08
5.5	53	9	13.4	40.2000	66.2	17	19.4	79.9882	0.08
6.0	52.7	15	13	50.3488	64.2	17	24.9	102.6653	0.108
6.5	45.8	12	23.1	80.0207	58.6	16	25.1	100.4000	0.179
7.0	48.4	15	12.7	49.1869	59.6	13	26.5	95.5471	0.183
7.5	38.7	14	22.1	82.6906	54.8	13	29.7	107.0849	0.122
8.0	38.6	18	21.3	90.3682	63.2	15	27.4	106.1197	0.007
8.5	29.5	11	17.5	58.0409	56.3	10	27.2	86.0140	0.014
9.0	26.5	14	14.7	55.0024	56	11	26.9	89.2172	0.005
9.5	30.9	12	18.3	63.3931	59.1	12	27.2	94.2236	0.008
10.0	26.8	10	18.5	58.5021	60.5	9	26	78.0000	0.004
10.5	31	8	16.5	46.6690	60.1	10	24.9	78.7407	0.013
11.0	29.5	7	20.1	53.1796	59.3	6	25.6	62.7069	0.038

Table 1. Mean BSID Cognition Raw Scores from Study 902 (n=32) and Study 201+Ext (n=22).

The most striking takeaway from Table 1 is that in almost every age bracket, the standard deviation is larger than the mean. In a neurotypical population, this would suggest impossible data. However, in MPS III-B, this reflects the extreme heterogeneity of the disease. In Study 902 (natural history), the variance explodes as children age, reflecting the reality that some children retain function longer while others crash early. In Study 201+Ext, the variance is even higher here, suggesting the population includes high-responders and those who may have started treatment too late to preserve existing cognitive architecture (cortical grey matter volume). Further, the mean differential (delta) between Study 902 and Study 201+Ext is shown in the table below.

Chronological Age	BSID Raw Score Mean Difference (Delta)	P-Value (Study 902 vs. Study 201/Ext)
4.0	16.82	0.111
4.5	6.60	0.55
5.0	14.90	0.08
5.5	13.20	0.08
6.0	11.50	0.108
6.5	12.80	0.179
7.0	11.20	0.183
7.5	16.10	0.122
8.0	24.60	0.007
8.5	26.80	0.014
9.0	29.50	0.005
9.5	28.20	0.008
10.0	33.70	0.004
10.5	29.10	0.013
11.0	29.80	0.038

Table 2. Mean BSID Raw Score Delta between Study 902 (n=32) and Study 201+Ext (n=22) by Age.

While the p-values in Table 1 suggest statistical significance, a p-value only indicates the probability that a difference exists; it does not measure the magnitude of that difference. To determine the ground-truth clinical impact, I calculated Cohen's d and applied the Hedges' g correction. The Hedges' g correction is critical here because it penalizes the effect size for the small sample sizes and unequal variances present in the cognition data presented. Without this correction, the effect of the drug would be mathematically overstated.

Chronological Age	Weighted Pooled Standard Deviation	Cohen’s d Effect Size	Hedges’ g Corrected
4.0	47.41767986	0.3547200126	0.3299721048
4.5	64.52790094	0.1022813373	0.09670235524
5.0	64.0627739	0.2325843714	0.2237520535
5.5	69.3115671	0.1904444028	0.184430369
6.0	82.48893259	0.1394126417	0.1358980372
6.5	92.32861878	0.1386352376	0.134597318
7.0	74.27130619	0.1507984789	0.1464062902
7.5	95.18333047	0.1691472648	0.1640215901
8.0	97.79648717	0.2515427774	0.2454075877
8.5	72.64662125	0.3689091046	0.3541527405
9.0	71.90739213	0.410249894	0.3967251722
9.5	80.30180571	0.3511751666	0.3390656781
10.0	68.37375658	0.4928791643	0.470809948
10.5	66.63656091	0.4366972065	0.4159021014
11.0	57.70553778	0.5164149082	0.4803859612

Table 3. Hedges’ g Corrected Effect Sizes. Standardized Effect Size Thresholds: 0.2 (Small), 0.5 (Medium), 0.8 (Large).

A Hedges’ g above 0.8 is considered a large effect. The calculated values for TA-ERT primarily hover between 0.13 and 0.48. From Ages 4 to 7.5, the effect sizes here are negligible (below 0.20), reinforcing the theory that the drug’s benefit is obscured by the developmental plateau in the natural history cohort. We see the effect size climb after Age 7.5. While this is a small-to-medium effect, it is important to contextualize this within a neurodegenerative cliff. In MPS III-B, maintaining a medium effect size is biologically significant. I would appreciate feedback on this point. Importantly, there is a notable tension between the sponsor’s p-values and these modest effect sizes. This discrepancy is the hallmark of high-variance data. A small group of super-responders in the treatment group can drive down a p-value in a MMRM model, but when we look at the standardized effect across the group, the clinical noise remains high. This suggests that while the drug is doing something statistically significant, the absolute clinical separation between the average treated child and the average untreated child remains narrow. This reinforces my neutral stance on the drug bending the curve but not being a “magic bullet” that completely separates the treatment population from the disease’s natural history.

While cross-sectional snapshots provide a state of the patient, the most significant metric for a neurodegenerative therapy is the Velocity of Decline, the rate at which cognitive function is lost over time. By calculating the first derivative of the mean BSID-III raw scores, we can filter out the noise of baseline heterogeneity and isolate the drug’s impact on the disease trajectory. The following table tracks the velocity for both Study 902 and Study 201+Ext.

Chronological-Age-Years	Time Elapsed	Study 902 Velocity	Study 201 Velocity	Velocity Difference
4.0				
4.5	0.5	8.64	-11.8	-20.44
5.0	0.5	1.6	18.2	16.6
5.5	0.5	9.8	6.4	-3.4
6.0	0.5	-0.6	-4	-3.4
6.5	0.5	-13.8	-11.2	2.6
7.0	0.5	5.2	2	-3.2
7.5	0.5	-19.4	-9.6	9.8
8.0	0.5	-0.2	16.8	17
8.5	0.5	-18.2	-13.8	4.4
9.0	0.5	-6	-0.6	5.4
9.5	0.5	8.8	6.2	-2.6
10.0	0.5	-8.2	2.8	11
10.5	0.5	8.4	-0.8	-9.2
11.0	0.5	-3	-1.6	1.4

Table 4. Velocity of BSID-III Raw Scores. A negative velocity indicates cognitive loss; a positive velocity difference indicates the treatment cohort is declining slower or stabilizing compared to natural history.

The velocity data identifies a critical 18-month window from Ages 7.5 to 9.0 that serves as the strongest proof-of-concept for TA-ERT. Between Ages 7.5 and 9.0, the untreated cohort experiences the cognitive cliff. During the same window, the TA-ERT cohort shows a less profound decline, yielding a consistently positive velocity difference. This is the biological bending of the curve. The drug appears to intercept the disease mechanism as the accumulation of substrate (HS-NRE) would otherwise lead to total cognitive failure.

When we isolate the 7.5 to 9.0 year window, the evidence for TA-ERT as a disease-modifying stabilizer is at its peak. During this phase, the Hedges' g effect size climbs from 0.16 to nearly 0.40, precisely as the velocity differential remains positive. This suggests TA-ERT is postponing the cognitive cliff associated with MPS III-B.

Beyond Age 9.0, the velocity difference becomes increasingly volatile. At first glance, the negative velocity differences at Age 9.5 and 10.5 suggest a narrowing of the treatment effect. However, an audit of the N-at-risk reveals a mathematical artifact known as Floor-Effect Censoring in the natural history cohort. For example, between Age 9.0 and 9.5, the untreated group's N dropped from 14 to 12, yet the total cognitive points for the group remained static (371.0 vs 370.8), calculated by multiplying the N-at-Risk by the mean score. This indicates that the children who dropped out were effectively at a zero-point floor. The removal of these low-performers artificially inflated the natural history mean, creating a statistically 'positive' velocity (+8.8) that is biologically impossible in untreated MPS III-B. In contrast, the treated cohort remained functionally stable (Mean ~60), well above the clinical floor, suggesting that the 'narrowing' effect is a byproduct of a corrupted natural history benchmark rather than a loss of drug efficacy.

To ensure the observed stabilization between ages 7.5 and 9.0 was not an artifact of cohort shifting, a Point-Pool attrition analysis (Mean*N-At-Risk) was conducted to determine the implied score of patients dropping out of the study. See table 5 below!

[**Important Note:** My Point-Pool Analysis and Velocity Analysis treats Study 902 and Study 201+Ext as apples-to-apples. I cannot verify that the analyses aren't partially driven by baseline imbalances. Accounting for the confounder of Baseline Heterogeneity requires patient-level covariate modeling, which Spruce uses in their MMRM model. Due to the black-box nature of their MMRM model, my analysis intends to provide insight while acknowledging the limitation of its applicability]

Chronological Age	Study 902 Total Points	Net Change in N	Net Change in Total Points	Implied Score of Shift	Study 201 Total Points	Net change in N	Net Change in Total Points	Implied Score of Shift
4.0	214.9				478.4			
4.5	283.8	1	68.9	68.9	539	2	60.6	30.3
5.0	481	4	197.2	49.3	756	2	217	108.5
5.5	477	-1	-4	4	1125.4	5	369.4	73.88
6.0	790.5	6	313.5	52.25	1091.4	0	-34	No N Change
6.5	549.6	-3	-240.9	80.3	937.6	-1	-153.8	153.8
7.0	726	3	176.4	58.8	774.8	-3	-162.8	54.27
7.5	541.8	-1	-184.2	184.2	712.4	0	-62.4	No N Change
8.0	694.8	4	153	38.25	948	2	235.6	117.8
8.5	324.5	-7	-370.3	52.9	563	-5	-385	77
9.0	371	3	46.5	15.5	616	1	53	53
9.5	370.8	-2	-0.2	0.1	709.2	1	93.2	93.2
10.0	268	-2	-102.8	51.4	544.5	-3	-164.7	54.9
10.5	248	-2	-20	10	601	1	56.5	56.5
11.0	206.5	-1	-41.5	41.5	355.8	-4	-245.2	61.3

Table 5. Point-Pool Attrition Analysis. Study 902 colored in yellow. Study 201+Ext colored in Blue. *Implied Score of Shift attributes all net point changes to the shifting N. This results in bloated implied scores during periods of high internal variance. *The purpose of this analysis is to prove the wild velocity swings seen in later ages is attributed to attrition in natural history. This analysis cannot be used to prove Study 201 did not just recruit healthier patients, which is why FDA requires MMRM Covariate Modeling.*

The analysis reveals a divergence in attrition mechanics between the two cohorts. In the Natural History group (Study 902), the mathematically positive velocity shifts at Age 9.5 and 10.5 (Table 4) are directly caused by Informative Censoring; the implied cognitive scores of the dropouts at these visits were 0.1 and 10.0 points, respectively. The removal of these floor-effect patients artificially inflated the untreated mean. Conversely, attrition in the TA-ERT treated cohort was driven by random censoring. When the treated cohort lost patients at Age 10 (n=-3) and Age 11 (n=-4), the implied scores of those dropouts were 54.9 and 61.3 points, squarely in line with the functionally stable group mean. The point-pool analysis shows that while the natural history cohort is artificially buoyed by the death or severe decline of its sickest patients, the TA-ERT cohort retains genuine functional stability. The Bending of the Curve observed in the 7.5 to 9.0 year window appears to be a true biological signal, untainted by the survivorship bias that plagues the later years of the control data.

Focusing specifically on Age 8 proves challenging. At Age 8, Study 201 had an observed velocity of 16.8, a net change in N of 2, and an implied score of shift of 117.8 (impossible as BSID-III maxes out at 91). To explain this, I think the +16.8 velocity is not a clean biological signal but the result of cohort swapping. When those two kids joined the cohort, they were likely at the top of the scale (near 91). The formula used to calculate implied score shift attributes all new points to the newcomers, thus the extra points (the difference between a perfect 91 and the 117.8 calculated) came from the internal gains of the original 13 kids. At Age 8, the treated cohort did not just add two healthy kids; the existing kids also had a period of temporary improvement or noise that boosted the mean. Now, at Age 8 the observed effect size was 0.25 (a small effect). The culprit of this is the weighted pooled standard deviation used in the Hedges' g Correction (97.8). When the spread is nearly 100 points, a mean difference in BSID score of 25 is statistically quiet, resulting in a small effect size despite high velocity. If it were me, I would want cleaner data. On a positive note, though Age 8 is clearly a data artifact caused by lucky recruitment, the cohort remained stable through age 9.0 while the natural history group continued to plummet (note: Age 9.5 and 10.5 attrition bias of the NH Cohort explains the positive velocity seen at those points).

My Final Thoughts

The objective of this analysis was to deconstruct the aggregate BSID-III cognition data to determine if TA-ERT demonstrates a clinically meaningful alteration of the MPS III-B disease trajectory. By auditing the underlying variance, correcting the standardized effect sizes, and isolating longitudinal velocity, this report identifies a biologically plausible mechanism of stabilization.

The Limits of Cross-Sectional Data: The reverse-calculated standard deviations reveal extreme heterogeneity within both cohorts, rendering point-in-time raw score comparisons statistically unreliable without heavy longitudinal modeling. The Hedges' g calculations confirm that the absolute magnitude of the treatment effect remains modest (g = 0.35 to 0.48 at older ages). TA-ERT does not uniformly restore patients to a neurotypical baseline; rather, the clinical separation is driven by a stabilization of decline.

The Stabilization Window: The first-derivative velocity analysis isolates the ages of 7.5 to 9.0 as the critical window of drug efficacy. During this period, the TA-ERT cohort seemingly resists the precipitous neurodegenerative cliff observed in the natural history population.

Informative Censoring: The point-pool attrition analysis mathematically argues that the apparent "narrowing" of the velocity differential after Age 9.0 is an artifact of Floor-Effect Censoring. The natural history cohort's late-stage velocity is artificially buoyed by the continuous dropout of 0-point patients, whereas the treated cohort's attrition reflects random censoring of functionally stable patients.

While this aggregate analysis successfully isolates the signal from the noise, it is inherently limited by the absence of patient-level data. The validity of the 7.5–9.0 stabilization window relies heavily on the assumption that the treatment and natural history cohorts were adequately matched at baseline. Because summary-level math cannot perform covariate adjustments or calculate within-subject covariance, the definitive regulatory proof of this stabilization rests within Spruce's unobservable MMRM data.

Assuming baseline comparability, I believe my analysis supports the thesis that TA-ERT successfully intercepts the neurodegenerative cascade, providing a clinically meaningful delay of cognitive decline that satisfies the requirement for a correlation between biomarker reduction and clinical benefit.

Remember that Spruce used an enrollment criteria of DQ>50. They account for this (and we will see if successful or not) with the MMRM model that adjusts for baseline DQ.

Thank you for reading,

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